(6) A problem in Mathematics is given to three students whose chances of solving it are $\frac{1}{3}, \frac{1}{4}$, and $\frac{1}{5}$ (i) What is the probability that the problem is solved? (ii) What is the probability that exactly one of them will solve it?

(7) The probability that a car being filled with petrol will also need an oil change is 0.30; the probability that it needs a new oil filter is 0.40; and the probability that both the oil and filter need changing is 0.15.

(i) If the oil had to be changed, what is the probability that a new oil filter is needed?

(ii) If a new oil filter is needed, what is the probability that the oil has to be changed?

(8) One bag contains 5 white and 3 black balls. Another bag contains 4 white and 6 black balls. If one ball is drawn from each bag, find the probability that (i) both are white (ii) both are black (iii) one white and one black

(9) Two thirds of students in a class are boys and rest girls. It is known that the probability of a girl getting a first grade is 0.85 and that of boys is 0.70. Find the probability that a student chosen at random will get first grade marks.

(10) Given $P(A) = 0.4$ and $P(A \cup B) = 0.7$. Find $P(B)$ if

(i) A and B are mutually exclusive (ii) A and B are independent events

(iii) $P(A / B) = 0.4$ (iv) $P(B / A) = 0.5$

(11) A year is selected at random. What is the probability that

(i) it contains 53 Sundays (ii) it is a leap year which contains 53 Sundays

(12) Suppose the chances of hitting a target by a person X is 3 times in 4 shots, by Y is 4 times in 5 shots, and by Z is 2 times in 3 shots. They fire simultaneously exactly one time. What is the probability that the target is damaged by exactly 2 hits?

12.7 Total Probability of an event

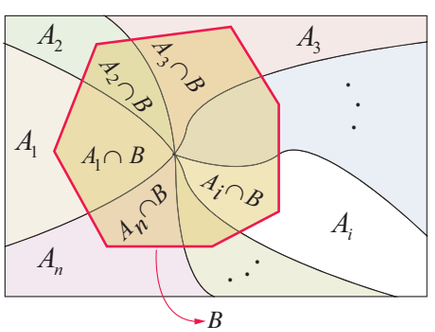
Theorem 12.10 (Total Probability of an event)

If $A_1, A_2, A_3, ..., A_n$ are mutually exclusive and exhaustive events and B is any event in S then $P(B)$ is called the total probability of event B and

$$P(B) = P(A_1).P(B/A_1) + P(A_2).P(B/A_2) + \cdots + P(A_n).P(B/A_n) = \sum_{i=1}^{n} P(A_i).P(B/A_i)$$

Proof

Since B is any event in S, from the figure shown here



$$B = (A_1 \cap B) \cup (A_2 \cap B) \cup (A_3 \cap B) \cup \cdots (A_n \cap B).$$

Since $A_1, A_2, A_3 \ldots A_n$ are mutually exclusive,

$(A_1 \cap B), (A_2 \cap B), (A_3 \cap B), \cdots, (A_n \cap B)$ are also mutually exclusive.

Therefore,

$$P(B) = P\left[(A_1 \cap B) \cup (A_2 \cap B) \cup (A_3 \cap B) \cup \cdots \cup (A_n \cap B)\right]$$

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B) + \cdots + P(A_n \cap B)$$

$$P(B) = P(A_1).P(B/A_1) + P(A_2).P(B/A_2) + \cdots + P(A_n).P(B/A_n) = \sum_{i=1}^{n} P(A_i).P(B/A_i).$$ ■

The following problems are solved using the law of total probability of an event.

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